

THE ELECTRIC MOTOR

1. What is an electric motor

An electric motor is a **machine** that transforms electrical energy into rotational mechanical energy. Electric motors are widely used to produce movement in toys, household appliances, transport vehicles, power tools, industrial machines, water pumps, etc.

An electric motor can also behave as an electrical energy generator when it is forced to rotate. This electricity generator is much cheaper and more durable than electrochemical batteries.

2. History of the electric motor

In the **1820s**, H. C. Ørsted and Michael Faraday discovered the basic principles of electromagnetism, which were necessary for building motors.

Between **1834 and 1838**, the first practical electric motor was developed. It was used to power a boat carrying twelve people in Saint Petersburg.

In **1866**, Werner von Siemens patented the dynamo, initiating the industrial production of electricity using direct current.

From **1880** onwards, direct-current electrical grids and power stations began to be built in many countries, including Spain.

In **1888**, Nikola Tesla built the first alternating-current motor. This form of current eventually became the one used in electrical distribution networks because of its advantages and because of the patents Tesla licensed free of charge to the Westinghouse company.

3. History of electrification in Spain

In Spain, the first company to produce and sell electricity (Sociedad Española de Electricidad) was founded in 1881 in Barcelona. However, it was not until many years later that electricity reached all households on a large scale.

Year	Energy generated	Households with electricity
1940	3.6 TWh	60 %
1950	6.9 TWh	80 %
1960	19 TWh	90 %
1970	57 TWh	95 %
1980	110 TWh	99 %
1990	152 TWh	100 %

4. Classification of motors

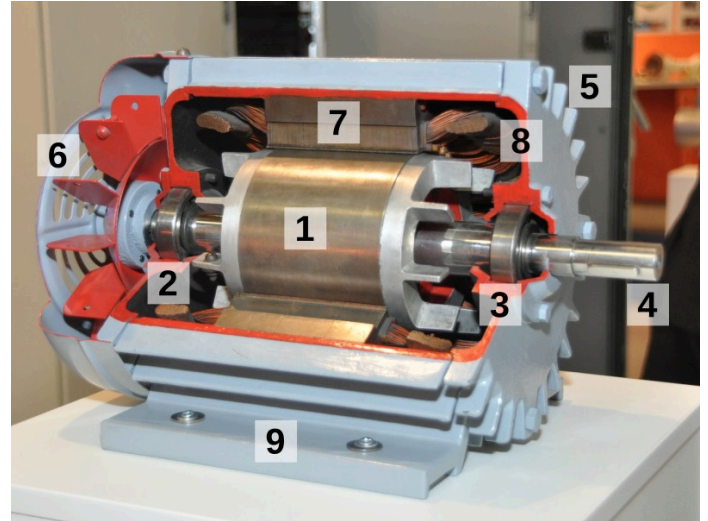
- Direct-current motors and universal motors.
- Synchronous motors and brushless motors.
- Induction motors.

- Reluctance motors and stepper motors.

5. Parts of the motor

An electric motor consists of two main components: the stator, which remains stationary, and the rotor, which rotates when the motor is operating.

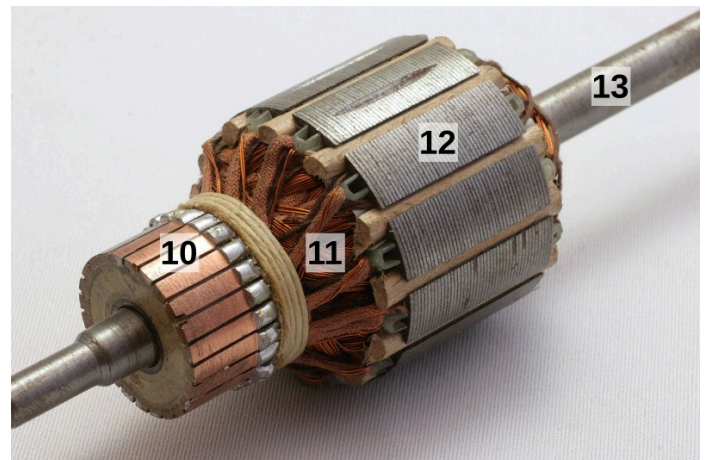
An open alternating-current induction motor, showing its interior:



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1. Squirrel-cage rotor (armature).
2. and 3. Bearings that support the rotor shaft.
4. Rotating shaft that transmits mechanical energy.
5. Housing with cooling fins.
6. Fan with blades that cools the housing.
7. Stator that generates a rotating magnetic field.
8. Stator coils supplied with alternating current.
9. Stator mounting foot used to secure the motor.

Rotor of a direct-current motor. In this motor, the stator's magnetic field is fixed:

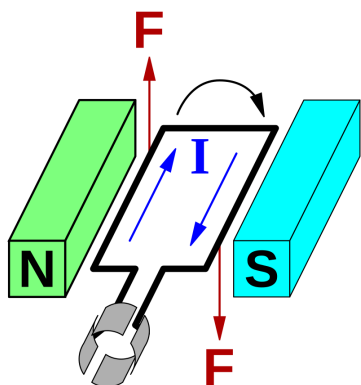


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10. Commutator with connection segments.
11. Copper wire winding (rotor coils).
12. Magnetic poles of the rotor.
13. Rotor shaft.

6. How an electric motor works

The operation of an electric motor is based on the force exerted by a magnetic field on an electric current (Lorentz force).



Direct-current motors have windings made of many insulated copper wires (11), through which current flows from the commutator (10). The stator's magnetic field is fixed, produced by permanent magnets or an electro-magnet.

The magnetic field generates a force on the current flowing through the copper wires, causing the rotor to rotate. If we reverse the direction of the current, the force also changes direction and the motor will rotate in the opposite direction.

As the rotor rotates, the commutator also rotates and supplies current to new rotor wires. In this way, the horizontal wires that produce the rotational force are always energized.

In **induction motors**, the rotor wires are replaced by conductive bars. The stator's magnetic field rotates and drags the rotor bars along with it.

7. The variable-frequency drive

A variable-frequency drive is an electronic device that controls the voltage and current supplied to the motor.



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The motor's **supply current** is proportional to the rotational force (torque). The **supply voltage**, and its frequency, are proportional to the motor's rotational speed. By controlling the current and voltage, the operation of the motor can be precisely controlled.

One application of a variable-frequency drive is to smoothly control the motors of vehicles so that they have constant acceleration. They can also control the speed of the means of transport.

When the variable-frequency drive is operating, it produces an audible buzzing sound that is characteristic of train motors and electric cars.

EXERCISES

1. What is an electric motor and what is it used for?
2. What is an electric generator and what is its relationship to motors?
3. Draw a timeline showing the main milestones in the history of the electric motor.
4. Draw a graph of the history of electrification in Spain. It should show one line representing the annual energy generated, with values on the left vertical axis in 15 TWh intervals, and another line representing the percentage of households with electricity, with values on the right vertical axis in 10% intervals.
5. Approximately what year did 85% of households in Spain have electricity installed?
6. Name 5 different types of electric motors.
7. Draw an induction motor and label its main parts.
8. Draw the rotor of a direct-current motor and label its main parts.
9. Explain how a direct-current motor works.
10. What is a variable-frequency drive for a motor, and what is it used for? Give an example of an application.
11. How can the rotational speed of a motor be controlled? And how can its torque be controlled?